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ASSIGNMENT: 6

1. **Write a Program to perform the following operations on the array:**
   a. Read Array
   b. Print Array
   c. Find Minimum Number from Array
   d. Find Maximum Number from Array

   Algorithm:
   1. Start
   2. Declare array and variables (n, min, max, i)
   3. Input array size n
   4. Loop from i = 0 to n-1:
      - Read a[i]
   5. Print array elements using a loop
   6. Initialize min = max = a[0]
   7. Loop from i = 1 to n-1:
      - If a[i] < min, update min
      - If a[i] > max, update max
   8. Print min and max
   9. End

   Code:

```
#include <stdio.h>

int main() {
   int a[100], n, i, min, max;

   printf("Enter size of array: ");
   scanf("%d", &n);

   printf("Enter elements:\n");
   for(i = 0; i < n; i++)
      scanf("%d", &a[i]);
```

```
    printf("Array elements: ");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);

    min = max = a[0];
    for(i = 1; i < n; i++) {
        if(a[i] < min) min = a[i];
        if(a[i] > max) max = a[i];
    }

    printf("\nMinimum = %d", min);
    printf("\nMaximum = %d", max);

    return 0;
}
```

2. **Write a program to perform the following operations on the array:**
   a. Read Array
   b. Print Array
   c. Search an Element from given array using linear search. Print the location of the element.

   Algorithm:
   1. Start
   2. Declare array and variables (n, key, found, i)
   3. Input size n
   4. Loop i = 0 to n-1:
      - Read a[i]
   5. Print array
   6. Input the element key to search
   7. Loop i = 0 to n-1:
      - If a[i] == key, set found = 1 and print position
      - Break loop
   8. If found == 0, print "Not found"
   9. End

   Code:

```
#include <stdio.h>

int main() {
   int a[100], n, i, key, found = 0;

   printf("Enter size of array: ");
   scanf("%d", &n);

   printf("Enter elements:\n");
   for(i = 0; i < n; i++)
      scanf("%d", &a[i]);

   printf("Array elements: ");
   for(i = 0; i < n; i++)
      printf("%d ", a[i]);
```

```
    printf("\nEnter element to search: ");
    scanf("%d", &key);

    for(i = 0; i < n; i++) {
        if(a[i] == key) {
            printf("Element found at position %d\n", i + 1);
            found = 1;
            break;
        }
    }

    if(!found)
        printf("Element not found.\n");

    return 0;
}
```

3. **Write a program to perform the following operations on the array:**
   a. Read Array
   b. Print Array
   c. Print sorted array using Bubble sort.

Algorithm:

1. Start
2. Declare array and variables (n, i, j, temp)
3. Input array size n and elements
4. Print original array
5. Loop i = 0 to n-2:
    - Loop j = 0 to n-i-2:
        - If a[j] > a[j+1], swap them
6. Print sorted array
7. End

Code:

```
#include <stdio.h>

int main() {
   int a[100], n, i, j, temp;

   printf("Enter size of array: ");
   scanf("%d", &n);

   printf("Enter elements:\n");
   for(i = 0; i < n; i++)
      scanf("%d", &a[i]);

   printf("Original Array: ");
   for(i = 0; i < n; i++)
      printf("%d ", a[i]);

   // Bubble Sort
   for(i = 0; i < n-1; i++) {
      for(j = 0; j < n-i-1; j++) {
```

```
            if(a[j] > a[j+1]) {
                temp = a[j];
                a[j] = a[j+1];
                a[j+1] = temp;
            }
        }
    }

    printf("\nSorted Array: ");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

4. **Write a Program to Perform the following operations on a 2D Array:**
   a. Addition of two matrices.
   b. Transpose of a matrix.
   c. Display diagonal elements of the matrix
   d. Display upper diagonal & lower diagonal elements of the matrix

Algorithm:

1. Start
2. Declare 2D arrays a, b, sum, trans and variables (i, j, n)
3. Input size n of square matrix
4. Input elements of matrix A and B
5. Addition:
   - Loop i = 0 to n-1, j = 0 to n-1:
     - sum[i][j] = a[i][j] + b[i][j]
6. Transpose:
   - Loop i = 0 to n-1, j = 0 to n-1:
     - trans[j][i] = a[i][j]
7. Diagonal Elements:
   - Loop i = 0 to n-1:
     - Print a[i][i]
8. Upper Diagonal:
   - Loop i = 0 to n-1, j = i+1 to n-1:
     - Print a[i][i]
9. Lower Diagonal:
   - Loop i = 1 to n-1, j = 0 to i-1:
     - Print a[i][j]
10. End

Code:

```
#include <stdio.h>

int main() {
   int a[10][10], b[10][10], sum[10][10], trans[10][10];
   int i, j, n;
```

```
printf("Enter size of square matrix (n x n): ");
scanf("%d", &n);

printf("Enter first matrix:\n");
for(i = 0; i < n; i++)
   for(j = 0; j < n; j++)
      scanf("%d", &a[i][j]);

printf("Enter second matrix:\n");
for(i = 0; i < n; i++)
   for(j = 0; j < n; j++)
      scanf("%d", &b[i][j]);

// Matrix Addition
printf("Addition of matrices:\n");
for(i = 0; i < n; i++) {
   for(j = 0; j < n; j++) {
      sum[i][j] = a[i][j] + b[i][j];
      printf("%d ", sum[i][j]);
   }
   printf("\n");
}

// Transpose
printf("Transpose of first matrix:\n");
for(i = 0; i < n; i++) {
   for(j = 0; j < n; j++) {
      trans[j][i] = a[i][j];
   }
}
for(i = 0; i < n; i++) {
   for(j = 0; j < n; j++)
      printf("%d ", trans[i][j]);
   printf("\n");
}

// Diagonal Elements
printf("Main Diagonal Elements: ");
for(i = 0; i < n; i++)
   printf("%d ", a[i][i]);
```

```
    printf("\nUpper Diagonal: ");
    for(i = 0; i < n; i++)
        for(j = i+1; j < n; j++)
            printf("%d ", a[i][j]);

    printf("\nLower Diagonal: ");
    for(i = 1; i < n; i++)
        for(j = 0; j < i; j++)
            printf("%d ", a[i][j]);

    return 0;
}
```

5. **Write a Program to Accept String from the user. Perform the following operations on the String (Without Using Built in Functions):**
   a. Print String Length
   b. Print String in reverse order
   c. Copy given string into other string and print both strings.
   d. Accept two strings from user and compare both. Print if they are equal. Or which one is greater or which one is small.
   e. Accept two strings from user and print its concatenated string.

   Algorithm:
   1. Start
   2. Declare strings: str1, str2, s1, s2, str3
   3. Input str1
   4. Length:
      - Loop through str1 until '\0', count characters
   5. Reverse:
      - Print characters from last to first index
   6. Copy:
      - Loop through str1, copy to str2
   7. Compare:
      - Input s1, s2.
      - Loop through both:
        - If characters differ, break and compare
   8. Concatenate:
      - Copy s1 to str3

- Append s2 at the end of str3

9. Print results
10. End

Code:

```
#include <stdio.h>

int main() {
    char str1[100], str2[100], str3[100];
    int i, len = 0, cmp = 0;

    printf("Enter a string: ");
    scanf("%s", str1);

    // Length
    while(str1[len] != '\0')
        len++;

    printf("Length = %d\n", len);

    // Reverse
    printf("Reverse = ");
    for(i = len - 1; i >= 0; i--)
        printf("%c", str1[i]);

    // Copy
    for(i = 0; i <= len; i++)
        str2[i] = str1[i];

    printf("\nOriginal: %s", str1);
    printf("\nCopied: %s\n", str2);

    // Compare two strings
    char s1[100], s2[100];
    printf("Enter first string: ");
    scanf("%s", s1);
    printf("Enter second string: ");
    scanf("%s", s2);

    for(i = 0; s1[i] != '\0' || s2[i] != '\0'; i++) {
```

```
        if(s1[i] != s2[i]) {
            cmp = s1[i] - s2[i];
            break;
        }
    }

    if(cmp == 0)
        printf("Strings are equal.\n");
    else if(cmp > 0)
        printf("First string is greater.\n");
    else
        printf("Second string is greater.\n");

    // Concatenation
    i = 0;
    while(s1[i] != '\0') {
        str3[i] = s1[i];
        i++;
    }
    int j = 0;
    while(s2[j] != '\0') {
        str3[i] = s2[j];
        i++;
        j++;
    }
    str3[i] = '\0';

    printf("Concatenated String: %s\n", str3);

    return 0;
}
```